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$$f' = f_d\left(1+\frac{v_0}{v}\right) \Rightarrow f_d = \frac{f'}{\left(1+\frac{v_0}{v}\right)} \qquad (11.92)$$

Since this is true for any arbitrary time, therefore, comparing equation (11.91) and equation (11.92), we get

$$\frac{f}{\left(1-\frac{v_s}{v}\right)} = \frac{f'}{\left(1+\frac{v_0}{v}\right)}$$

$$\Rightarrow \frac{v f'}{(v+v_0)} = \frac{v f}{(v-v_s)}$$

Hence, the apparent frequency as seen by the observer is

$$f' = \left(\frac{v+v_0}{v-v_s}\right) f \qquad (11.93)$$

(b) Source and observer recede from each other

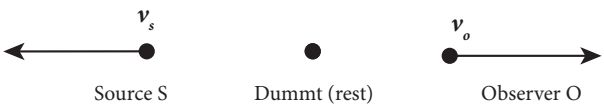


Figure 11.49 Source and Observer resides from each other

Here, we can derive the result as in the previous case. Instead of a detailed calculation, by inspection from Figure 11.49, we notice that the velocity of the source and the observer each point in opposite directions with respect to the case in (a) and hence, we substitute ($v_s \rightarrow -v_s$) and ($v_0 \rightarrow -v_0$) in equation (11.93), and therefore, the apparent frequency observed by the observer when the source and observer recede from each other is

$$f' = \left(\frac{v-v_0}{v+v_s}\right) f \qquad (11.94)$$

(c) Source chases the observer

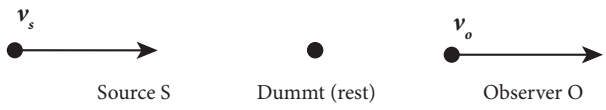


Figure 11.50 Source chases observer

Only the observer's velocity is oppositely directed when compared to case (a). Therefore, substituting ($v_0 \rightarrow -v_0$) in equation (11.93), we get

$$f' = \left(\frac{v-v_0}{v-v_s}\right) f \qquad (11.95)$$

(d) Observer chases the source

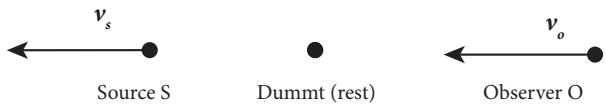


Figure 11.51 Observer chases Source

Only the source velocity is oppositely directed when compared to case (a). Therefore, substituting $v_s \rightarrow -v_s$ in equation (11.93), we get

$$f' = \left(\frac{v+v_0}{v+v_s}\right) f \qquad (11.96)$$

Discuss with your teacher

"Doppler effect in light"

"Doppler effect in sound is asymmetrical where as Doppler effect in light is symmetrical"

11.11.4 Applications of Doppler effect

Doppler effect has many applications. Specifically Doppler effect in light has many applications in astronomy. As an example, while observing the spectra from distant